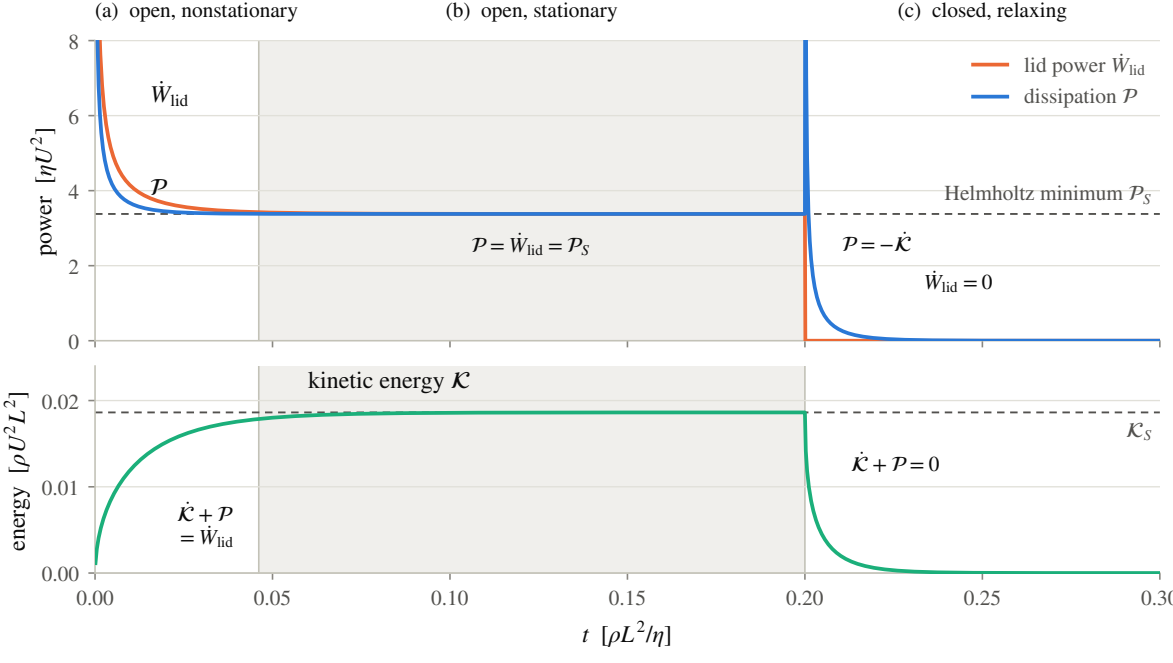
(a) open, nonstationary
(b) open, stationary
(c) closed, relaxing
lid power $\dot{W}_{\text{lid}}$
dissipation $\mathcal{P}$
power $[\eta U^2]$
$\dot{W}_{\text{lid}}$
$\mathcal{P}$
Helmholtz minimum $\mathcal{P}_S$
$\mathcal{P} = \dot{W}_{\text{lid}} = \mathcal{P}_S$
$\mathcal{P} = -\dot{\mathcal{K}}$
$\dot{W}_{\text{lid}} = 0$
kinetic energy $\mathcal{K}$
energy $[\rho U^2 L^2]$
$\mathcal{K}_S$
$\dot{\mathcal{K}} + \mathcal{P} = \dot{W}_{\text{lid}}$
$\dot{\mathcal{K}} + \mathcal{P} = 0$
t $[\rho L^2/\eta]$